Which of the following is an example of Matrix in reduced Echelon form?

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

Question # 1 of 10 (Start time: 07:46:22 PM, 01 June 2022) Total Marks: 1

If $\alpha = \beta$, then $\begin{vmatrix} Cos\alpha & -Sin\alpha \\ Cos\beta & Sin\beta \end{vmatrix} =$

Select the correct option

Reload Math Equations

- Undefined
- (∞)
- $Sin2\alpha$
- $Cos2\alpha$
- 1

Question # 3 of 10 (Start time: 07:46:55 PM, 01 June 2022)

Inverse of a matrix is given by

Select the correct option

- $A^{-1} = \frac{1}{detA} A^{-1}$
- $A^{-1} = \frac{1}{detA} adj(A)$

Question # 2 of 10 (Start time: 07:47:00 PM, 01 June 2022)

If

$$A = \begin{bmatrix} 2 & -7 \\ 4 & 3 \end{bmatrix}$$

, then adjoint of A is

Select the correct option

- $$A = \begin{bmatrix} 2 & 7 \\ -4 & 3 \end{bmatrix}$$
- $$A = \begin{bmatrix} 3 & 7 \\ -4 & 2 \end{bmatrix}$$

Question # 4 of 10 (Start time: 07:47:31 PM, 01 June 2022)

How many Pivot positions the matrix : $\begin{pmatrix} 2 & 3 & 1 \\ 4 & 6 & 2 \end{pmatrix}$ *will have?*

Select the correct option

- 2
- 4
- 1
- 3

Question # 3 of 10 (Start time: 07:48:03 PM, 01 June 2022)

An n×n real matrix is invertible if and only if the span of the rows of A is R^n

Select the correct option

- True
- False

Question # 4 of 10 (Start time: 07:48:45 PM, 01 June 2022)

The transpose of the product of two matrices is equal to

Select the correct option

○	$(AB)^T = A^T B^T$
○	$(AB)^T = B^T A^T$

Question # 5 of 10 (Start time: 07:48:34 PM, 01 June 2022)

The Elementary Row operations: $R_2' \to R_1 + 4R_1$ and $R_3' \to R_3 - 6R_1$ are performed on to get $\begin{pmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{pmatrix} \sim$ - - - - - - - ?

Select the correct option

- $\begin{pmatrix} 1 & 2 & -5 \\ 0 & 9 & 26 \\ 0 & -9 & -26 \end{pmatrix}$
- $\begin{pmatrix} 1 & 2 & -5 \\ 0 & -9 & 26 \\ 0 & 9 & -26 \end{pmatrix}$
- $\begin{pmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & -9 & 26 \end{pmatrix}$
- $\begin{pmatrix} 1 & 2 & -5 \\ 0 & -9 & -26 \\ 0 & -9 & 26 \end{pmatrix}$

Question # 5 of 10 (Start time: 07:49:30 PM, 01 June 2022)

Linear algebra is the study of linear sets of equations and their ---------- properties.

Select the correct option

- transformation
- non-linear

$$\begin{pmatrix} Cos\theta & Sin\theta \\ -Sin\theta & Cos\theta \end{pmatrix}^{-1} =$$

$$\begin{pmatrix} Cos\theta & Sin\theta \\ Sin\theta & -Cos\theta \end{pmatrix}$$

$$\begin{pmatrix} Cos\theta & Sin\theta \\ -Sin\theta & Cos\theta \end{pmatrix}$$

$$\begin{pmatrix} Cos\theta & -Sin\theta \\ Sin\theta & Cos\theta \end{pmatrix}$$

$$\begin{pmatrix} -Cos\theta & Sin\theta \\ Sin\theta & Cos\theta \end{pmatrix}$$

Question # 5 of 10 (Start time: 07:50:01 PM, 01 June 2022)

If

$$A = \begin{bmatrix} 2 & -7 \\ 4 & 3 \end{bmatrix}$$

, then adjoint of A is

Select the correct option

- $A = \begin{bmatrix} 2 & 7 \\ -4 & 3 \end{bmatrix}$
- $A = \begin{bmatrix} 3 & 7 \\ -4 & 2 \end{bmatrix}$

Question # 7 of 10 (Start time: 07:50:45 PM, 01 June 2022)

Algebraic expression involving more than one term and less than 3 terms, is known as

Select the correct option

- Binomial
- Monomial
- Multinomial
- Trinomial

The following Equations:

$$2x - y = 3$$

$$-x + \frac{1}{2}y = \frac{-3}{2},$$

is an example of - - - - - System.

Select the correct option

- Homogeneous
- Inconsistent
- Non-Linear
- Consistent

Question # 8 of 10 (Start time: 07:51:39 PM, 01 June 2022)

If the sum of two matrices A, and B is a zero matrix then A and B are said to be

Select the correct option

O	Transpose of each other.
O	Additive inverse of each other.
O	Determinant of each other.
	Multiplicative inverse of each other.

Question # 9 of 10 (Start time: 07:52:16 PM, 01 June 2022)

The equation $Ax = b$ has a solution if and only if

Select the correct option

○	b is the linear combination of the columns of A
○	b is a non-linear combination of the columns of A

Question # 8 of 10 (Start time: 07:52:04 PM, 01 June 2022)

The transpose of the product of two matrices is equal to

Select the correct option

- $(AB)^T = B^T A^T$
- $(AB)^T = A^T B^T$

Question # 10 of 10 (Start time: 07:52:58 PM, 01 June 2022)

$The\ matrix : [x_{p1}]\ , where\ 1 \leqslant p < \infty, is\ an\ example\ of\ - - - -matrix.$

Select the correct option

- Column
- Row
- Square
- Singular

Question # 9 of 10 (Start time: 07:53:01 PM, 01 June 2022)

Which of the following can refer a single term of an Algebraic expression?

Select the correct option

- 2x-3y
- 6xy
- (6x/y) such that y=0
- 2x+3y

Which of the following is Row - Equivalent of $\begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$?

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

$\begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$

$\begin{pmatrix} 2 & -4 \\ -1 & 3 \end{pmatrix}$

$\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$

Question # 1 of 10 (Start time: 07:55:19 PM, 01 June 2022)

For the matrix: $A = \begin{pmatrix} 4 & x+2 \\ 2x-3 & 1 \end{pmatrix}$, if $A = A^t$, then $x = ---$.

Select the correct option

- Undefined.
- 5
- -5
- 5/2

Question # 2 of 10 (Start time: 07:56:09 PM, 01 June 2022)

If $\alpha = \beta$, then $\begin{vmatrix} Cos\alpha & -Sin\alpha \\ Cos\beta & Sin\beta \end{vmatrix} =$

Select the correct option

○	$Cos2\alpha$
○	Undefined (∞)
○	1
○	$Sin2\alpha$

Under which of the following condition, a system of Linear Equations whose Row - Reduce form is $\left(\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 0 & h-3k \end{array}\right)$ has *Infinite* many solutions?

$(h,k) = (0,0)$

$(h,k) \neq (0,0)$

$h = 3k$

$h \neq 3k$

Question # 4 of 10 (Start time: 07:57:47 PM, 01 June 2022)

If A and B are matrices then A+B is only possible if both matrices have same ------

Select the correct option

○	Elements
○	Order
○	Rows
○	Columns

Question # 5 of 10 (Start time: 07:58:40 PM, 01 June 2022)

An n×n real matrix is invertible if and only if the span of the rows of A is R^n

Select the correct option

- False
- True

Question # 6 of 10 (Start time: 07:59:10 PM, 01 June 2022)

We can subtract two matrices A and B if their-------

Select the correct option

○	Order is same
○	Rows are same
○	Elements are same
○	Columns are same

Question # 7 of 10 (Start time: 08:00:15 PM, 01 June 2022)

In a matrix, vertically arranged elements are called

Select the correct option

- Transpose
- Determinants
- Rows
- Columns

Question # 8 of 10 (Start time: 08:01:37 PM, 01 June 2022)

$$\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} \times \begin{bmatrix} 8 & -4 & 5 \end{bmatrix} =$$

Select the correct option

○	$\begin{bmatrix} 24 \\ -8 \\ -5 \end{bmatrix}$
○	$\begin{bmatrix} 24 & -8 & -5 \end{bmatrix}$
○	$\begin{pmatrix} 24 & -12 & 15 \\ 16 & -8 & 10 \\ -8 & 4 & -5 \end{pmatrix}$
○	11

Question # 9 of 10 (Start time: 08:02:30 PM, 01 June 2022)

The transpose of the product of two matrices is equal to

Select the correct option

- $(AB)^T = B^T A^T$
- $(AB)^T = A^T B^T$

Question # 10 of 10 (Start time: 08:03:10 PM, 01 June 2022)

Let A, B and C matrices are of same order and $A(B+C)=AB+AC$, this law is known as

Select the correct option

○	Associative law
○	Commutative law
○	Distributive law
○	Cramer's law

A matrix whose all entries are zero is called ------

Select the correct option

- a)Unit matrix
- (a) and (b)
- b)Null matrix
- c)Identity matrix

Which of the following will be the Matrix Product corresponding to Linear Combination $\begin{pmatrix} -2 \\ 1 \end{pmatrix} x + \begin{pmatrix} 3 \\ 1 \end{pmatrix} y$?

correct option

$\begin{pmatrix} -2 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} 3 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} 3 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} -2 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

Which of the following is corresponding Matrix form of the Linear equation $x + y = 3$?

on

ressed in Matrix form:

$$x\begin{pmatrix}1\\0\end{pmatrix}+y\begin{pmatrix}0\\1\end{pmatrix}=3\begin{pmatrix}1\\0\end{pmatrix}$$

$$x\begin{pmatrix}1\\0\end{pmatrix}+y\begin{pmatrix}0\\1\end{pmatrix}=3\begin{pmatrix}0\\1\end{pmatrix}$$

$x[1] + y[1] = 3[1]$

Which of the following is true about the existence of free variables (parameter) in a system of Linear Equations?

lect the correct option

- They do not guarantee the Consistency.
- They guarantee the Consistency.
- None of above all
- They guarantee the Inconsistency.

if the sum of two matrices A and B is a zero matrix then A and B are said to be

Select the correct option

- Transpose of each other.
- Multiplicative inverse of each other.
- Determinant of each other.
- Additive inverse of each other.

In a matrix, vertically arranged elements are called

- Determinants
- Rows
- Transpose
- Columns

The non - zero values of x and y which satisfy the matrix equation: $x\begin{pmatrix}2\\y\end{pmatrix}-y\begin{pmatrix}1\\-2\end{pmatrix}$

option

$\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}2\\-4\end{pmatrix}$

$\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}-2\\-4\end{pmatrix}$

$\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}-4\\-2\end{pmatrix}$

$\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}4\\-2\end{pmatrix}$

$$\begin{vmatrix} \int_0^1 e^x dx & \int_0^1 e^{1-x} dx \\ -\int_0^1 e^{1-x} dx & \int_0^1 e^x dx \end{vmatrix} =$$

Select the correct option

- $2(e-1)^2$
- $e-1$
- $(e-1)^2$
- 0

Linear algebra is the study of linear sets of equations and their ---------- properties.

elect the correct option

- non-linear
- transformation

The Equation: $2x - 3y + kz = 4$, is linear in ------.

x, y, z and k

any one variable

x, y and z

x and y only

The following Equations:

$$2x - y = 3$$

$$-x + \frac{1}{2}y = \frac{-3}{2},$$

is an example of - - - - - - System.

---

the correct option

- Non-Linear
- Homogeneous
- Inconsistent
- Consistent

If $\begin{pmatrix} 2x & 8y+4x \\ 8 & -7 \end{pmatrix} = \begin{pmatrix} 16 & 0 \\ 8 & -7 \end{pmatrix}$, then $(x, y) = ---$,

ct the correct option

- (8,-4)
- (-4,8)
- (-8,4)
- (4,-8)

The symbols or letters that are used to represent a quantity that varies, are referred to as ——————

Select the correct option

- None of the above
- Polynomials
- Variables
- Constants

The equation $Ax = b$ has a solution if and only if

elect the correct option

- b is the linear combination of the columns of A
- b is a non-linear combination of the columns of A

Which of the following would be the value of t, if $\begin{bmatrix} 2e^{-3t} \\ 5e^{-3t} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

e correct option

$\pm\frac{1}{3}$

zero

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\frac{\begin{bmatrix} 2 \\ 5 \end{bmatrix}}{\begin{bmatrix} 2 \\ 5 \end{bmatrix}} = 1$

Which of the following will be the Matrix Product corresponding to Linear Combination: $\begin{pmatrix} -2 \\ 5 \end{pmatrix} x + \begin{pmatrix} 5 \\ 1 \end{pmatrix} y$?

rrect option

$\begin{pmatrix} -2 & 5 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} 5 & -2 \\ 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} -2 & 5 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} 1 & -5 \\ -5 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

If a matrix is in reduced row echelon form, then it is also in row echelon form.

Select the correct option

- May be
- True
- False
- None of the above